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Epidemics
Editorial Board

Dear Editors,

We are pleased to submit our manuscript titled “Evaluation of stochastic trajectory-based epidemic models using the energy score” by Bay et al. for consideration in *Epidemics*.

The importance of rigorous evaluation frameworks for epidemic forecasts and scenario projections has grown significantly as multi-team collaborative efforts, such as the Scenario Modeling Hub and the CDC’s FluSight Challenge, have become essential for informing public health decisions. This work introduces the energy score as a well-suited performance metric for evaluating stochastic trajectory-based epidemic models, addressing limitations in current scoring methodologies for multivariate time-series epidemic predictions.

In this paper, we present the energy score, a multivariate extension of the continuous ranked probability score (CRPS), for evaluating epidemic projections in stochastic trajectory format. More specifically we provide the following contributions:

- We compare the energy score with WIS, demonstrating its ability to distinguish between forecasts that have the same marginal distribution but differ in trajectory structure.
- We illustrate how the energy score improves evaluation of complex, multi-peak epidemic models by overcoming the limitations of model communication using fixed-time descriptive statistics (e.g. quantiles) in capturing the full variability of model trajectories.
- We apply the energy score to the 2023–2024 Flu Scenario Modeling Hub projections, illustrating how this metric performs across multiple scenarios and models submitted by various teams.
- We introduce a normalized energy score to assess multi-target outcomes (e.g., cases, hospitalizations, deaths, geographical locations) in a unified metric, improving interpretability for public health decision-making.

We believe these contributions are timely, given the increasing adoption of ensemble epidemic forecasts and the need for evaluation frameworks that accommodate trajectory-based epidemic models. Our study provides practical insights for improving predictive analytics in public health, with implications for epidemic forecasting and scenario modeling hubs. For this reason, we view *Epidemics* as the ideal venue to share these findings and engage the scientific community most invested in methodological innovation for epidemic modeling and evaluation.

Thank you very much for your consideration.

Sincerely, on behalf of all the authors,

Clara Bay